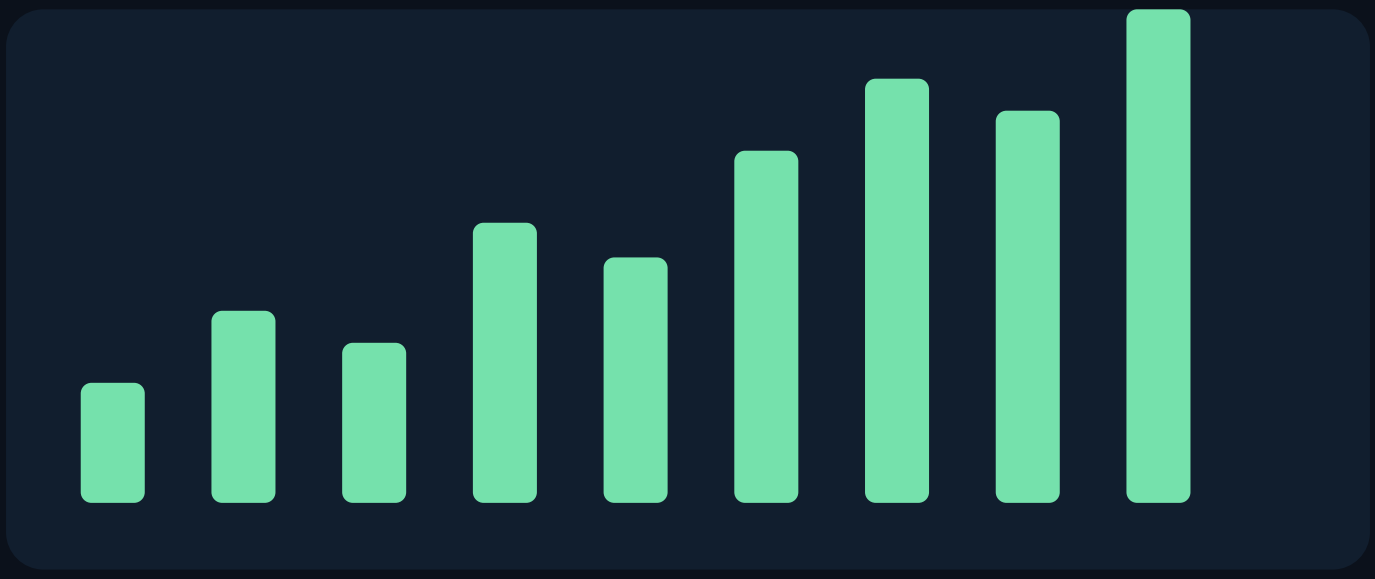
PRACTICAL KNOWLEDGE / DARK EDITION

Advanced Excel & MIS Reporting

Build a monthly reporting workbook that refreshes cleanly, explains its numbers and survives handover.

By Datta Sable



DATTA SABLE / PROFESSIONAL FIELD GUIDES

THE READER BRIEF

A guide you can put to work

A professional MIS workbook is a small reporting system. It has an input contract, a transformation process, a calculation layer and an output that helps someone decide what to do. Colour and charts are useful only after those layers agree.

This guide develops a six-row sales example into a controlled monthly brief. Every expected total is supplied, including a deliberately blank region. You will practise formulas, exception handling, Power Query design, PivotTables, presentation and refresh control.

Build the workbook as you read. Keep raw input, calculations, checks and presentation on different sheets. The goal is a workbook another analyst can refresh without manually repairing formulas or guessing which cells are safe to edit.

What you need

Excel for Microsoft 365 on desktop for the full workflow. XLOOKUP, dynamic arrays, Power Query and refresh interfaces vary by version and platform. Formulas use English names and comma argument separators; adapt separators to your regional settings.

Edition and scope

Original educational examples by Datta Sable. First edition, 28 October 2026. This purchase contains this PDF only. Code, tables and exercises are printed inside; no separate datasets, software licences or editable files are bundled. Product interfaces and availability can change. Check the linked official documentation when applying a platform-specific step.

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YOUR LEARNING ROUTE

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FIELD GUIDE / 01

Design the workbook as a reporting system

Create five sheets: ReadMe, Input, Mapping, Checks and Report. Add a sixth, Calculations, when formulas become too complex to audit on the report itself. Keep raw source values intact; an unexplained manual correction is invisible technical debt.

A small contract

Decide which date controls the reporting month, whether amounts include tax, how returns appear and what a blank region means. In this guide, Amount is paid merchandise value excluding tax. All six rows are valid paid transactions. Blank Region means unclassified and must remain visible in a quality check.

Useful controls

Put the report month in Report!B2 as a real date such as 1 January 2026, displayed as mmm yyyy. Put a refresh timestamp and source filename on ReadMe. Distinguish input cells with a subtle fill and a short label, rather than relying on colour alone. Protect formulas against accidental editing, while retaining a documented recovery path.

Handoff test

Ask a colleague to identify the source table, month selector, refresh command and exception list in under a minute. If they cannot, improve labels before adding more calculations. A workbook can be technically correct and still be operationally fragile.

Your deliverable

Write a five-line ReadMe: purpose, source, owner, refresh sequence and known limitations. Include the Excel version used. This becomes the baseline for every later chapter.

FIELD GUIDE / 02

Enter and validate the practice table

Paste the following CSV values into Input, then use Insert > Table and name the table Sales. Confirm that Date contains Excel dates, Units and Amount are numeric, and OrderID is text. Do not include a total row inside the raw data.

```
OrderID,Date,Region,Product,Units,Amount
S001,2026-01-03,West,Guide,2,198
S002,2026-01-05,South,Template,1,149
S003,2026-01-10,West,Guide,3,297
S004,2026-01-15,North,Course,1,499
S005,2026-02-02,South,Guide,1,99
S006,2026-02-04,,Template,2,298
```

Baseline answers

There are six rows, ten units and 1,540 in amount. January contributes 1,143 across four rows and seven units. February contributes 397 across two rows and three units. The unclassified region contributes 298. Keep those numbers on Checks before building the dashboard.

Type problems are data problems

An ISO date string can remain text depending on import settings. Changing its number format does not convert its value. Check ISNUMBER for a sample date or import with an explicit type and locale through Power Query. Leading zeros in identifiers can be lost if the column is inferred as numeric.

Exercise and answer

Temporarily type Amount as text on one row. SUM may ignore it while a visual inspection still sees digits. A numeric-count check should detect the mismatch. Restore the value and record the lesson: formatting is not validation.

FIELD GUIDE / 03

Use structured references and explicit periods

Excel Tables expand as new rows are added and make formulas easier to read. Sales[Amount] says more than F2:F5000 and does not depend on guessing a future row limit. Use real dates to construct reporting windows rather than comparing formatted month names.

```
Monthly amount:
=SUMIFS(Sales[Amount],Sales[Date],">="&$B$2,
 Sales[Date],"<"&EDATE($B$2,1))

Monthly units:
=SUMIFS(Sales[Units],Sales[Date],">="&$B$2,
 Sales[Date],"<"&EDATE($B$2,1))

Monthly transactions:
=COUNTIFS(Sales[Date],">="&$B$2,
 Sales[Date],"<"&EDATE($B$2,1))
```

The line breaks above are for readability; enter each expression as one formula. With B2 set to 1 January 2026, the answers are 1,143, 7 and 4. EDATE calculates the next month's boundary. The less-than comparison prevents February 1 from joining January's total.

Freeze assumptions, not every cell

B2 remains fixed when a formula is copied. Structured references remain attached to table columns. Use named cells such as ReportMonth when that improves readability, and document them in Name Manager. Excessive unnamed helper cells make handover harder.

Practice and answer

Change B2 to February 1. The results must become 397, 3 and 2 without modifying any source row or formula. If they do not, inspect data types and date boundaries first.

FIELD GUIDE / 04

Look up mappings without hiding exceptions

Suppose product labels need a reporting category. Create a table named ProductMap with Product and Category columns: Guide maps to Learning, Template to Tools and Course to Learning. Each product key must appear once.

```
=XLOOKUP([@Product],ProductMap[Product],
 ProductMap[Category],"UNMAPPED",0)
```

Add that formula to a calculated Category column in Sales. The explicit exact-match setting communicates intent. UNMAPPED is an actionable state; replacing all missing matches with an empty string would make incomplete classification hard to see.

Duplicate mapping keys

XLOOKUP returns a matching result, not a warning that the mapping table has duplicate keys. Add a count check beside each mapping key:

```
=COUNTIF(ProductMap[Product],[@Product])
```

Every mapping row should return 1. If a key appears twice with different categories, resolve the conflict before refreshing the report. Do not assume that whichever row happens to appear first is authoritative.

Compatibility note

Older Excel installations may not support XLOOKUP. An exact INDEX/MATCH combination can provide the lookup, but keep the same missing-key and duplicate-key checks. Test the workbook in the recipient's actual environment rather than depending on a cached displayed result.

Exercise and answer

Change one product to Bundle. The category should read UNMAPPED and your exception count should increase by one. Restore the original after confirming the control works.

FIELD GUIDE / 05

Build a visible quality-control panel

Checks should be independent enough to challenge the report. A second SUM of the same range is a reconciliation, but it will not detect a bad source value if both formulas accept it. Combine structural checks, business-rule checks and reconciled totals.

```
Input rows:
=ROWS(Sales[OrderID])

Blank regions:
=COUNTBLANK(Sales[Region])

Numeric amount count:
=COUNT(Sales[Amount])

Per-row duplicate flag:
=COUNTIF(Sales[OrderID],[@OrderID])>1

Per-row validity:
=AND(ISNUMBER([@Date]),ISNUMBER([@Amount]),
 [@Amount]>=0,[@Units]>0)
```

For the supplied table, blank regions equals 1 and numeric amount count equals 6. A blank region is a warning in this exercise, not grounds to delete a legitimate sale. Keep an Unclassified bucket on the regional report so totals still reconcile.

Reconcile the complete population

Regional totals should sum to 1,540, including Unclassified. Monthly totals should also sum to 1,540. If your regional chart totals 1,242, it has silently dropped the 298 blank-region transaction.

Make failures actionable

Show check name, expected result, actual result, status and the affected row count. Add a note that identifies the person who can correct the source. Avoid a single green "all good" cell whose logic nobody can inspect.

Exercise and answer

Duplicate S001 in a new row. The duplicate flag should activate on both occurrences. A quality summary that counts flagged rows will show two affected rows, not one unique duplicated identifier; label that distinction.

FIELD GUIDE / 06

Plan a refreshable Power Query pipeline

Power Query is useful when recurring files need the same cleaning steps. Design the transformation as a reproducible recipe: connect, choose the intended source object, preserve useful lineage, set types, normalise agreed labels, validate and load.

A controlled sequence

1. Connect to a copied practice file or table.
2. Inspect column names before promoting headers.
3. Keep OrderID as text; set Date with the intended locale.
4. Trim surrounding whitespace in Region and Product.
5. Preserve blank Region as null; classify it explicitly downstream.
6. Add SourceFile when combining files.
7. Load to a table or model with a documented destination.

Do not use Remove Errors as a general cleaning strategy. It can reduce the report population without leaving a visible exception trail. Separate invalid rows into an exceptions query and reconcile accepted plus rejected rows to the source count.

Folder imports need boundaries

A folder query can ingest last month's output, temporary lock files or an old copy of the same input. Filter filenames and extensions before combining, and exclude the workbook's own generated output. Record the source period so duplicate batches can be detected.

Exercise and answer

Create a practice copy with a text value in Amount. The intended outcome is a visible type error or exception row, not a lower sales total. After correction, row count and 1,540 source total must match the original baseline. A successful refresh message alone is not proof of complete data.

FIELD GUIDE / 07

Use PivotTables to investigate, not just decorate

Create a PivotTable from Sales. Put Region in Rows and Amount in Values, explicitly set to Sum. Add Date grouped to an appropriate month/year level where supported. Retain the blank-region row or replace it with a clearly named Unclassified value in a controlled transformation.

Reconcile the first view

West: 495.
South: 248.
North: 499.
Unclassified: 298.
Grand total: 1,540.

If the Values field shows Count of Amount, the column may contain text or the summary function may have been selected incorrectly. Changing the displayed number format will not fix a count.

Refresh and filters

A PivotTable does not always refresh simply because the source table changed. Include Refresh All in the runbook, then verify both the source table and PivotTable totals. Slicers make exploration visible, but a screenshot can conceal an active filter. Put the selected period and key filter state next to the chart.

Explainable drilldown

Use a detailed transaction view to investigate a surprising total. Any extracted detail is a snapshot; do not edit it and expect the source to change. Keep source corrections in the authorised input system or a documented adjustments table.

Exercise and answer

Filter to January. West remains 495, South becomes 149 and North remains 499. There is no January Unclassified transaction. Their sum must be 1,143. Clear the filter and confirm the total returns to 1,540.

FIELD GUIDE / 08

Calculate KPIs with honest denominators

A KPI is a formula plus a definition. Average transaction amount is monthly amount divided by monthly transaction count. Average unit value divides by units instead. Neither should be labelled simply "average sale" without a definition.

```
Average transaction amount:
=IF(TransactionCount=0,"",MonthlyAmount/TransactionCount)

Month-on-month growth:
=IF(PriorAmount=0,"",CurrentAmount/PriorAmount-1)
```

January average transaction amount is 285.75. February is 198.50. February revenue change relative to January is approximately -65.27%. With only two months of invented data, this is a calculation exercise, not a business trend.

Zero, blank and not applicable

If the prior period is zero, percentage growth is undefined under this formula. A blank or N/A with an explanatory note is more honest than infinity or 0%. IFERROR around every KPI can conceal broken references, so handle known denominator states explicitly and let unexpected formula errors remain visible.

Weighted aggregation

The combined average transaction amount is 1,540 / 6 = 256.67 rounded to two decimals. Averaging 285.75 and 198.50 gives 242.125, which incorrectly weights months equally despite different transaction counts.

Exercise and answer

What is the combined average amount per unit? 1,540 / 10 = 154.00. Keep unit value and transaction value in separate KPI cards so the audience understands which operational change each can reveal.

FIELD GUIDE / 09

Design a premium report that remains readable

Use visual hierarchy to answer three questions: what happened, why it changed and what needs action. A practical one-page layout has a period label at the top, three or four KPI cards, one time view, one category view and a small exceptions panel.

Dark presentation rules

Choose a deep navy or charcoal background, a slightly lighter card fill and one accent colour. Use near-white text for key values and a softer neutral for labels. Avoid pure neon text across entire paragraphs. Do not encode favourable and unfavourable states using colour alone; add signs, words or icons.

Match the chart to the question

A sorted horizontal bar chart makes regional comparison clear. A line chart supports a sequence of months, but two observations offer limited trend evidence. A table is better when exact values and exception descriptions matter. Avoid 3D effects, heavy borders and multiple unrelated scales.

Give numbers context

Show the reporting period, currency convention, last refresh and whether data is provisional. Round displayed amounts consistently while keeping underlying calculations at full precision. A note such as "Unclassified region: 1 transaction, 298" makes the quality issue visible without discarding revenue.

Exercise and answer

Remove chart decoration one element at a time. If the business question remains answerable, that decoration was optional. Print or export a preview to check clipped labels, tiny fonts and missing legends before sending the workbook.

FIELD GUIDE / 10

Operate, protect and hand over the workbook

A refresh runbook should be short enough to use and specific enough to detect failure. Record where the source comes from, who owns it, the refresh order, expected checks, output location and what to do when a check fails.

A repeatable monthly run

Save the received source unchanged with its reporting date. Confirm file type and period. Refresh queries, then PivotTables. Compare row counts and monetary totals with source evidence. Review exceptions, update the commentary and export the management view. Save a dated copy of the workbook and the delivered PDF together.

Protect the right things

Sheet protection can reduce accidental edits; it is not a substitute for file access control. Keep confidential data in an approved location and share with the intended audience. Remove unused source extracts from a distributed workbook when recipients should only receive the report, and verify what remains in hidden sheets and query connections.

Acceptance test

A second analyst should change the reporting month, add a legitimate row, refresh and explain the resulting totals. For our sample, adding a January transaction of 100 increases January to 1,243 and the grand total to 1,640. No formula range should require manual expansion.

Final deliverable

Submit the workbook, metric definitions, six-row baseline, exception evidence and one-page report. The measure of professionalism is not the number of formulas. It is whether the next refresh produces a complete, traceable answer with minimal intervention.

KEEP LEARNING

Reference desk

Official references checked while preparing this edition. Use these pages for syntax, feature availability and the latest product details. The exercises and business scenarios in this book are original.

Microsoft: XLOOKUP function

https://support.microsoft.com/en-us/excel/functions/xlookup-function

Microsoft: SUMIFS function

https://support.microsoft.com/en-us/office/sumifs-function-c9e748f5-7ea7-455d-9406-611cebce642b

Microsoft: Power Query documentation

https://learn.microsoft.com/en-us/power-query/

Microsoft: Excel Tables and structured references

https://support.microsoft.com/en-us/office/using-structured-references-with-excel-tables-f5ed2452-2337-4f71-bed3-c8ae6d2b276e

Make the next step observable

Build the six-row workbook and hand it to another person with only your ReadMe. Observe where they hesitate. Improve the workbook until they can refresh, reconcile and explain it without verbal instructions.

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